

SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1718

COURSE OUTCOME COVERED

CO4: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 20%

Time Duration: 1 Hour

QUESTIONS: 5 Total Marks:30

Annexure A

ARTICLE REVIEW

Code of Conduct:

I SANJAI S(192511018) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: SANJAI S

CSA1718 AI
CO4 AT2 ARTICLE REVIEW

RESEARCH ARTICLE REVIEW

Statistical Learning Methods – Support Vector Machine and Naïve Bayes

ARTICLE SELECTED

ARTICLE TITLE:

CLOUD-BASED EMAIL PHISHING ATTACK USING MACHINE AND DEEP LEARNING ALGORITHM

AUTHORS:

UMER AHMED BUTT, RASHID AMIN, HAMZA ALDABBAS, SENTHILKUMAR MOHAN, BADER ALOUFFI, AND ALI AHMADIAN

Journal: ***Complex & Intelligent Systems, Springer***

Volume: 9

Pages: 3043–3070

Publication: 2022; Volume issue published in 2023

DOI: 10.1007/s40747-022-00760-3

Publisher: Springer Nature

Article Type: Original Research Paper / Peer-reviewed

Application Domain: Cybersecurity and cloud-based email security

Machine Learning Problem: Binary classification of emails into phishing/spam and legitimate/non-phishing categories.

The article was published by Springer and is openly accessible. ([Springer Link](#))

Article

Link:

[Springer – Full Research Article](#)

DOI

Link:

<https://doi.org/10.1007/s40747-022-00760-3>

TASK 1 – ARTICLE SUMMARY

1(a) Domain and Specific ML Problem

The research belongs to the cybersecurity domain, specifically email phishing detection. Phishing emails attempt to deceive users into revealing sensitive information such as passwords, financial information, or account credentials. Therefore, automatically identifying whether an email is phishing or legitimate is an important binary classification problem.

The authors investigate machine-learning-based email classification using Support Vector Machine (SVM), Naïve Bayes (NB), and Long Short-Term Memory (LSTM). SVM and Naïve Bayes are particularly relevant to CO4 because they are supervised/statistical learning techniques used for classification.

The authors collected spam and legitimate email data and extracted useful information from email headers, body text, and other email components. The extracted information was converted into a CSV-based dataset and processed using NLP techniques before classification. ([Springer Link](#))

1(b) Key Objective and Research Gap

Article Summary – approximately 180 words

The main objective of the research is to develop an effective machine-learning-based approach for detecting phishing emails and distinguishing them from legitimate emails. The authors identify email phishing as a significant cybersecurity problem because attackers use deceptive messages to obtain sensitive user information. Existing approaches often rely on limited content-based features or individual classification techniques, which may not adequately capture different characteristics of phishing emails. The research therefore attempts to improve detection by extracting multiple features from email headers, body text, and related email information and by comparing different machine-learning and deep-learning algorithms. The authors construct a modified dataset from more than 4,000 email files and convert the extracted information into a structured CSV format. NLP-based preprocessing and feature selection are then applied before classification. SVM, Naïve Bayes, and LSTM are evaluated for phishing detection. The research reports that SVM achieves the highest accuracy of 99.62%, compared with 97% for Naïve Bayes and 98% for LSTM. The proposed approach demonstrates that carefully selected email features combined with supervised learning can provide highly accurate phishing detection. ([Springer Link](#))

TASK 2 – METHODOLOGY ANALYSIS

2(a) ML Techniques Used

The paper evaluates three main algorithms:

1. Support Vector Machine (SVM)

SVM is a supervised statistical learning algorithm that attempts to find an optimal decision boundary between classes.

For this research, the classes are:

- Class 0 → Legitimate / non-phishing
- Class 1 → Phishing / spam

The paper uses a linear kernel, which is particularly suitable for high-dimensional text classification. The authors explain that the linear kernel is useful when a dataset contains a large number of features. ([Springer Link](#))

The basic idea is:

Email features → SVM → Decision boundary → Phishing / Legitimate

SVM is the primary CO4 technique in this review.

2. NAÏVE BAYES

Naïve Bayes is a probabilistic statistical learning method based on Bayes' theorem.

It estimates:

$$\frac{P(\text{Class}|\text{Features})}{P(\text{Class})}$$

and assigns the email to the class having the highest posterior probability.

Naïve Bayes is widely used for text classification because it is computationally simple and works effectively with high-dimensional textual features. The authors use it as another statistical learning benchmark. ([Springer Link](#))

3. Long Short-Term Memory (LSTM)

LSTM is a recurrent neural-network-based deep learning model.

It is used to process sequential information in email text and capture relationships between words.

Unlike SVM and Naïve Bayes, LSTM is a deep learning technique rather than a traditional statistical learning method.

The paper therefore provides a useful comparison between:

- Statistical learning → SVM
- Probabilistic learning → Naïve Bayes
- Deep learning → LSTM

Dataset Description

The authors collected spam and legitimate email data from the CSDMC_SPAM online dataset and additionally modified/combined email data.

The paper reports that the resulting collection contained more than 4,000 .eml email files. ([Springer Link](#))

Dataset characteristics

Parameter	Description
Domain	Email phishing/spam detection
Dataset source	CSDMC_SPAM and additional email data
Approximate size	4,000+ email files
File format	.eml
Target	Phishing/spam vs legitimate
Data type	Email text and metadata
Class distribution	Approximately 69% non-spam and 31% spam
Processing	Feature extraction, cleaning, NLP, feature selection

Parameter	Description
Output format	CSV + label file

The authors state that approximately 69% of the dataset was non-spam and 31% was spam. ([Springer Link](#))

Feature Extraction and Preparation

The processing pipeline can be represented as:

Raw .eml files

↓

Email header/body extraction

↓

Feature extraction

↓

Data cleaning

↓

Text processing / NLP

↓

Feature selection

↓

CSV dataset + labels

↓

Train/Test split

↓

SVM / Naïve Bayes / LSTM

↓

Classification result

The authors specifically mention extracting information from email headers, body, text, and connections. The extracted email files were converted into CSV format and labels were assigned to distinguish spam/phishing from legitimate messages. ([Springer Link](#))

2(b) Mapping to CO4 Topics

CO4 Topic	Connection with Article
Inductive Learning	The models learn classification rules from labelled examples of phishing and legitimate emails.
Decision Trees	Decision-tree-based approaches are discussed in the comparative analysis, although SVM/NB/LSTM are the primary evaluated techniques.

CO4 Topic	Connection with Article
Statistical Learning	SVM and Naïve Bayes directly represent statistical/supervised learning methods.
Reinforcement Learning	Not directly used in this research.
Classification	The core problem is binary classification.
Feature Selection	Irrelevant email characteristics are removed before classification.
NLP	Email text is processed to extract useful textual information.

Most Relevant CO4 Topic

Statistical Learning Methods – SVM and Naïve Bayes

The strongest connection is to statistical learning because the research uses SVM and Naïve Bayes to learn a decision function/probabilistic classification rule from labelled training data.

Methodological Strength

Strength: Multiple Algorithms and Feature Extraction

A major strength is that the authors do not depend on only one classifier. They compare SVM, Naïve Bayes, and LSTM and also perform feature extraction and preprocessing before classification.

This allows the researchers to determine which learning technique performs best for their email dataset. The SVM achieved a reported accuracy of 99.62%, exceeding the reported NB and LSTM results. ([Springer Link](#))

Another strength is the use of information from different parts of an email rather than depending exclusively on a single feature type.

Methodological Limitation

Limitation: Limited Generalization and Dataset Transparency

The reported accuracy is extremely high, but the study uses a relatively small dataset of only 4,000+ emails compared with the enormous diversity of real-world email traffic.

More importantly, the paper does not provide enough detail about the exact final number of samples used for every experiment, the complete feature list, hyperparameter settings, and an independent external test set.

The authors themselves suggest that future work should combine phishing and legitimate email data into a new dataset that more closely represents real-world conditions. ([Springer Link](#))

Therefore, the reported 99.62% accuracy should not automatically be interpreted as 99.62% performance on real-world emails.

TASK 3 – RESULTS AND CRITICAL EVALUATION

3(a) Key Results

The paper reports the following main classification results:

Algorithm	Accuracy	Precision	Recall	F-measure
SVM	99.62%	0.996	0.996	0.996
LSTM	98%	0.980	0.980	0.980
Naïve Bayes	97%	0.970	0.970	0.970
Random Forest	—	0.945	0.945	0.945
Logistic Regression	—	0.939	0.939	0.939
Neural Network	—	0.950	0.950	0.950

The paper reports SVM as the best-performing method, with approximately 99.62% accuracy, precision, recall and F-measure of approximately 0.996. ([PubMed Central \(PMC\)](#))

The reported training loss values were approximately:

Model	Accuracy	Loss
SVM	0.996	0.01
Naïve Bayes	0.97	0.03
LSTM	0.98	0.02

([Springer Link](#))

3(b) Critical Evaluation

Are the Results Realistic?

The reported 99.62% SVM accuracy is impressive, but it should be interpreted cautiously.

For a controlled dataset, such a result is possible. However, real-world phishing detection is significantly more difficult because attackers continuously modify:

- Email wording
- Sender addresses
- Domains
- URLs
- Attachments
- HTML structure
- Social-engineering techniques
- Obfuscation techniques

Therefore, a model achieving 99.62% on one dataset may perform substantially worse when evaluated on completely new emails.

Potential Dataset Bias

One potential bias is that the dataset is based primarily on a specific collection of spam/phishing emails.

The authors report that approximately 69% of the data is non-spam and 31% is spam. ([Springer Link](#))

Although this imbalance is not extreme, it means accuracy alone does not provide the complete picture.

For example, in cybersecurity, a false negative can be more serious than a false positive.

A false negative means:

A phishing email is incorrectly classified as legitimate.

This could result in:

- Credential theft
- Financial loss
- Malware infection
- Data leakage
- Organizational compromise

Therefore, recall, precision, F1-score, ROC-AUC, PR-AUC, and false-negative rate should all be carefully considered.

Evaluation Protocol Concern

Another concern is whether the test data is sufficiently independent from the training data.

If emails with similar templates, vocabulary, sender patterns, or duplicated information occur in both training and testing sets, the model may achieve artificially high performance.

This is particularly important for text classification.

Better evaluation would include:

1. Stratified train/test splitting.
 2. Five- or ten-fold cross-validation.
 3. An entirely independent external dataset.
 4. Time-based evaluation.
 5. Evaluation on newer phishing emails.
 6. Testing against deliberately obfuscated phishing messages.
-

Additional Experiments That Would Strengthen the Research

The following experiments would make the findings significantly stronger:

Experiment 1 – Cross-Dataset Validation

Train the SVM using CSDMC_SPAM and test it on another publicly available phishing dataset.

This would test whether the model generalizes beyond its original dataset.

Experiment 2 – Time-Based Testing

Train the model on older emails and test it on newer phishing emails.

This would simulate real-world deployment.

Experiment 3 – Ablation Study

Remove one feature group at a time:

- **Header features**
- **Body features**
- **URL features**
- **Text features**
- **Connection features**

Then measure how much each feature group contributes.

Experiment 4 – Compare More Statistical Models

Compare:

- **SVM**
- **Naïve Bayes**
- **Logistic Regression**
- **Decision Tree**
- **Random Forest**
- **Gradient Boosting**

This would create a stronger CO4-focused comparison.

Experiment 5 – Robustness Testing

Modify phishing emails using:

- **Spelling variations**
- **Synonyms**
- **URL obfuscation**
- **HTML changes**
- **Character substitutions**

Then evaluate whether the classifier still detects the phishing emails.

3(c) Real-World Applicability

The proposed approach can be deployed in several real-world environments.

1. Enterprise Email Security

Companies can integrate the classifier into their email gateway to automatically flag suspicious messages.

2. Cloud Email Services

Cloud-based email platforms can perform classification before messages reach the user's inbox.

3. Banking

Banks can use phishing detection to protect customers from fraudulent emails.

4. Educational Institutions

Universities can filter phishing messages targeting students, faculty, and administrative staff.

5. Government Organizations

Government email systems can use automated detection to reduce credential theft and malware distribution.

Challenges in Industrial Deployment

1. Data Availability

A large volume of accurately labelled phishing emails is required.

2. Concept Drift

Phishing techniques continuously change. A model trained today may become less effective later.

3. Privacy

Email content can contain confidential and personally identifiable information.

Therefore, organizations must carefully handle email data.

4. False Positives

If legitimate emails are incorrectly classified as phishing, important business communication may be blocked.

5. False Negatives

A missed phishing email can cause serious security damage.

6. Computational Requirements

SVM is relatively efficient for many text-classification applications, but feature extraction and large-scale email processing can become computationally expensive.

7. Adversarial Attacks

Attackers may deliberately modify email text to bypass machine-learning models.

TASK 4 – REFLECTION AND LEARNING OUTCOME

4(a) Three Key Insights

Insight 1 – Machine Learning Learns from Examples

The first important insight is that supervised machine learning can learn a classification boundary from labelled examples rather than requiring every phishing rule to be manually programmed.

In this paper, emails are labelled as phishing/spam or legitimate, and the algorithms learn from these examples.

CO4

Connection:

Inductive Learning

This demonstrates inductive learning because the model derives a general classification rule from previously observed training examples.

Insight 2 – Feature Selection Strongly Influences Classification

The second insight is that the quality of input features can strongly influence the final performance of a machine-learning model.

The authors extract information from email headers, body text and other email characteristics before applying classification. ([Springer Link](#))

CO4

Connection:

Statistical Learning / Feature Selection

A powerful algorithm cannot compensate completely for poor or irrelevant input features. Proper feature extraction can make classification more effective.

Insight 3 – Accuracy Alone Is Not Enough

The third insight is that a very high accuracy does not necessarily guarantee real-world reliability.

For phishing detection, recall and false-negative rate are particularly important because incorrectly allowing a phishing email to reach the user can have serious consequences.

The paper reports precision, recall and F-measure in addition to accuracy, which gives a more informative evaluation than accuracy alone. ([Springer Link](#))

CO4

Connection:

Statistical Model Evaluation

This helped me understand that machine-learning models must be evaluated using multiple metrics rather than relying only on accuracy.

4(b) Proposed Novel Extension

Proposed Experiment: Time-Aware Cross-Dataset Phishing Detection Using SVM

If I were extending this research, I would develop a time-aware cross-dataset SVM phishing detection system.

Instead of randomly dividing one dataset into training and testing data, the model would be trained using older phishing and legitimate emails and tested using newer emails from a different dataset.

Proposed methodology

Dataset A – Historical emails

↓

Feature extraction

↓

NLP preprocessing

↓

SVM training

↓

Dataset B – Newer real-world emails

↓

Independent testing

↓

Accuracy + Precision + Recall + F1 + ROC-AUC + PR-AUC

This experiment would measure whether the model can handle concept drift, where attackers change their email patterns over time.

Why is it feasible?

SVM is computationally efficient and works well with high-dimensional text data. The article already demonstrates the effectiveness of SVM for email classification, so the proposed extension can build directly upon the existing methodology. ([Springer Link](#))

Expected Impact

The proposed experiment would provide a more realistic estimate of real-world performance.

If accuracy decreases significantly on newer data, it would demonstrate the importance of continuous model retraining.

The system could therefore be extended into an adaptive phishing-detection system that periodically learns from newly identified phishing emails.

Overall Critical Assessment

The article provides a useful application of statistical learning methods to cybersecurity, particularly demonstrating the effectiveness of SVM and Naïve Bayes for email phishing classification. The strongest reported result is the SVM accuracy of 99.62%, with approximately 0.996 precision, recall and F-measure. ([PubMed Central \(PMC\)](#))

The major strength is the combination of feature extraction, NLP preprocessing, and comparison of multiple learning algorithms. However, the extremely high performance should be interpreted cautiously because real-world phishing emails are more diverse and continuously evolving.

The most important improvement would be to perform independent cross-dataset and time-based validation. Such experiments would determine whether the model has genuinely learned general phishing characteristics or has primarily learned patterns specific to the training dataset.

Overall, the research provides a strong practical example of Inductive Learning and Statistical Learning, especially SVM and Naïve Bayes, and demonstrates how CO4 concepts can be applied to an important real-world cybersecurity problem.

Conclusion

The reviewed article demonstrates how machine-learning algorithms can be applied to detect phishing emails automatically. The research compares SVM, Naïve Bayes, and LSTM, with SVM producing the highest reported accuracy of 99.62%. The study demonstrates important CO4 concepts such as supervised learning, inductive learning, statistical classification, feature selection, and performance evaluation.

The research is particularly valuable because phishing attacks continuously threaten individuals and organizations. Although the reported results are promising, additional validation using independent, newer, and more diverse datasets would be necessary before claiming that the system is ready for large-scale industrial deployment.

The proposed time-aware cross-dataset experiment could improve the reliability and generalization of the model and provide a more realistic evaluation of its effectiveness against continuously evolving phishing attacks.

Reference

Butt, U. A., Amin, R., Aldabbas, H., Mohan, S., Alouffi, B., & Ahmadian, A. (2023). *Cloud-based email phishing attack using machine and deep learning algorithm*. *Complex & Intelligent Systems*, 9, 3043–3070. <https://doi.org/10.1007/s40747-022-00760-3>. ([Springer Link](#))

Springer

Article:

[Cloud-based email phishing attack using machine and deep learning algorithm – Springer](#)

DOI:

[10.1007/s40747-022-00760-3](https://doi.org/10.1007/s40747-022-00760-3)